

Contextualized Generative Ghosting Model as Benchmark for Evaluating Player Movement in Football

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About Me

- ▶ I am a Ph.D. student at Loughborough University. My research is mainly about how to use machine learning methods to analyze big data in soccer to gain insights that can be used to help coaching staff make better tactical decisions.

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Introduction

- ▶ Current analytical models in soccer:

- ▶ Xg mode (Rathke, 2017)
- ▶ Expected Possession Value model
(Fernández, Bornn, and Cervone, 2019)
- ▶ VAEP model (Decroos et al., 2021)

- ▶ Generative Ghosting model:

- ▶ Features:

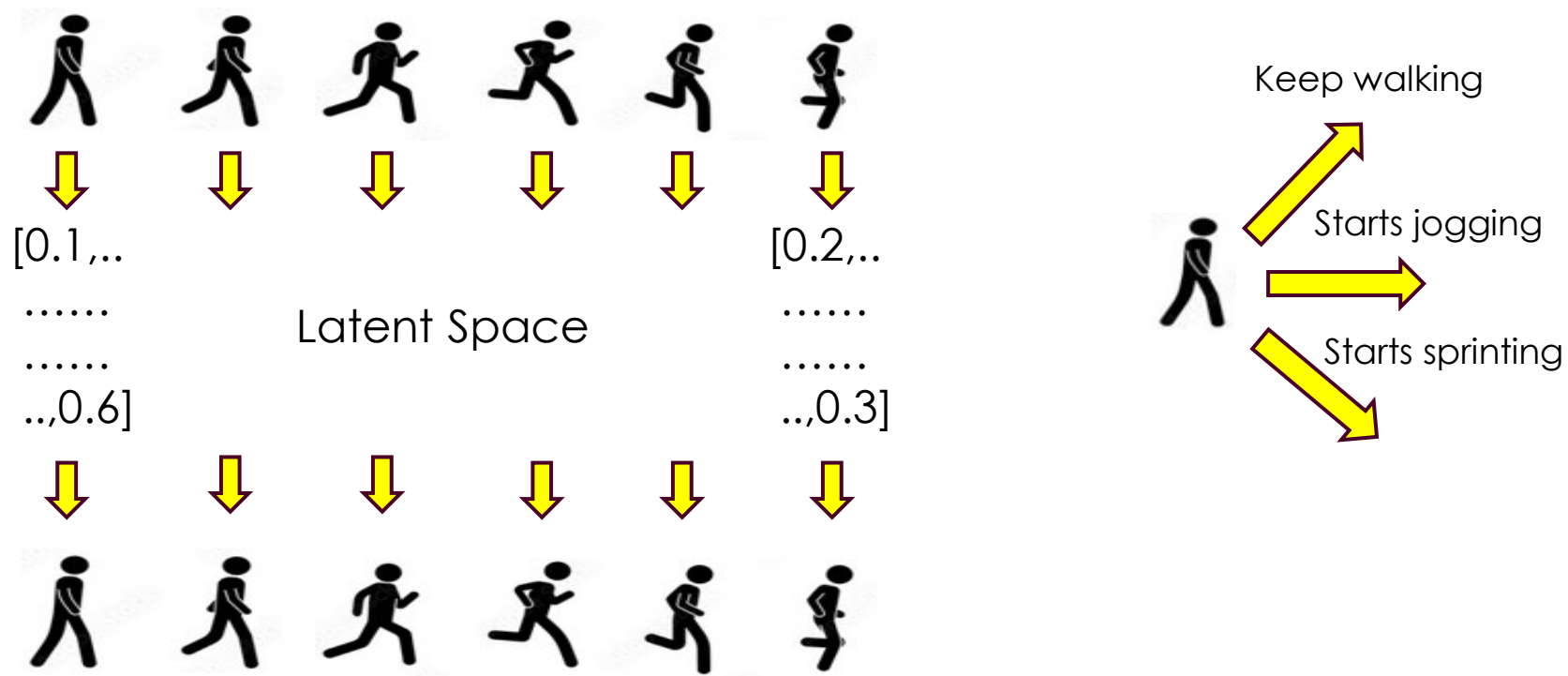
- ▶ Supervised learning with defined input and output
- ▶ Model performance heavily relies on the distribution of training data
- ▶ Cannot answer what-if questions in given context

- ▶ Features:

- ▶ Combination of unsupervised learning and supervised learning
- ▶ More likely to capture the generative patterns within the data
- ▶ Can answer what-if questions in given context

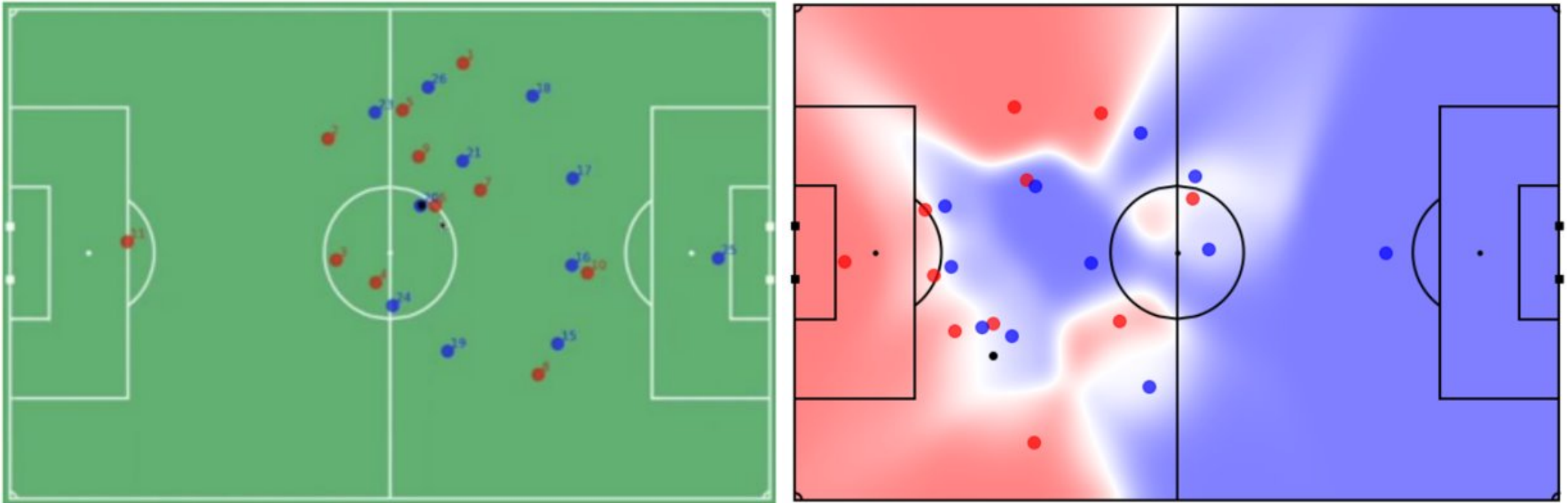
Background

- Deep Generative Model on Time Series Data (Chung et al., 2015)



Background

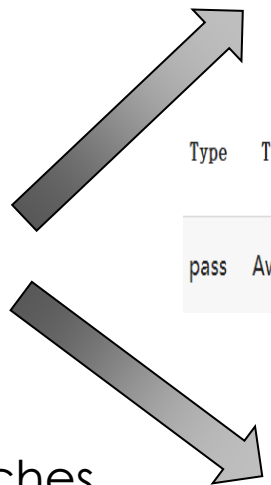
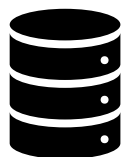
- Pitch Control (Spearman W, 2018)



Data



Professional
Soccer Matches



Event Data

Type	Team	From	To	Start Time [s]	End Time [s]	Ball_Start	Ball_End	Start X	Start Y	End X	End Y	Start Frame	End Frame
pass	Away	184667	95658	4.00	6.08	[2.7, 20.25, 8.59, 0.25]	[5.37, 24.02, -10.37, 0.24]	20.25	8.59	24.02	-10.37	100	152



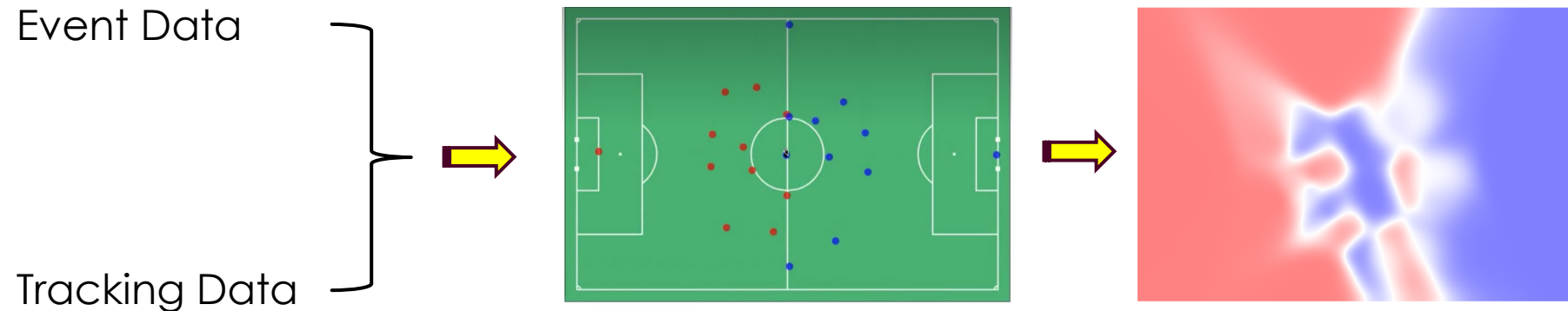
Tracking Data

Match_Phase	Time [s]	Match_Active	Last_Touch	Home_0_x	Home_0_y	...	Away_0_x	Away_0_y	...	ball_x	ball_y
1	0.00	False	home	-17.07	-8.40	...	21.46	8.98	...	0.75	0.40

Data Processing

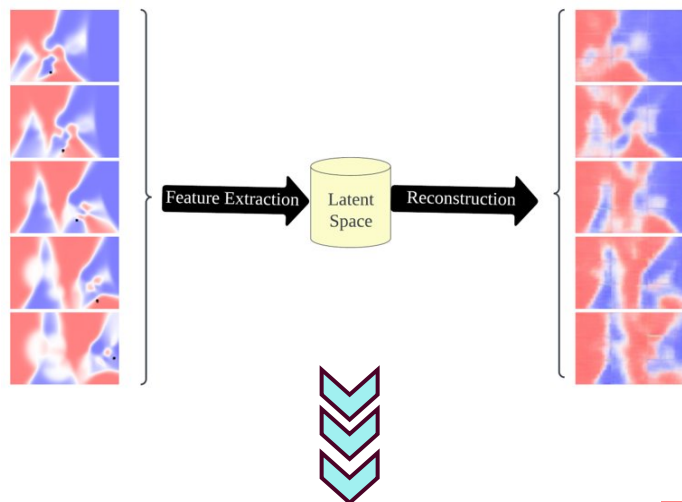
► Change data representation

Instead of directly analyzing tracking and event data, we convert them to another form of representation – pitch control frames

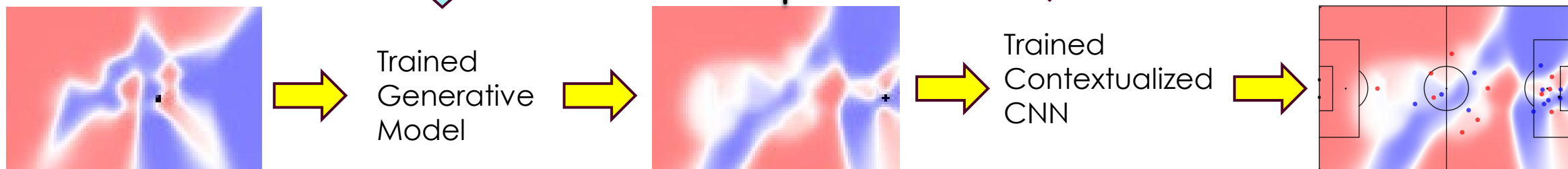


Framework of Generative Ghosting

Part 1

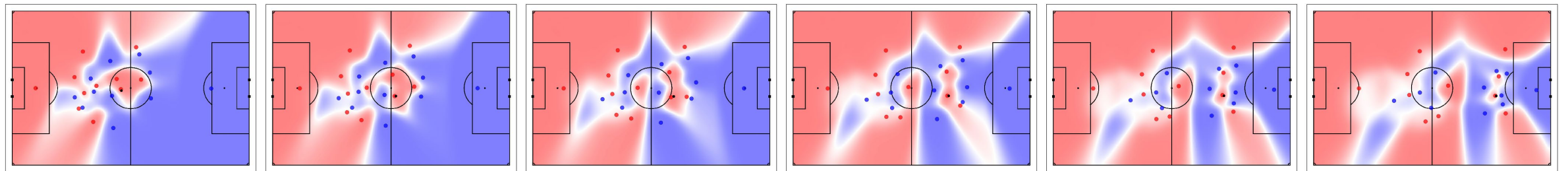


Part 2

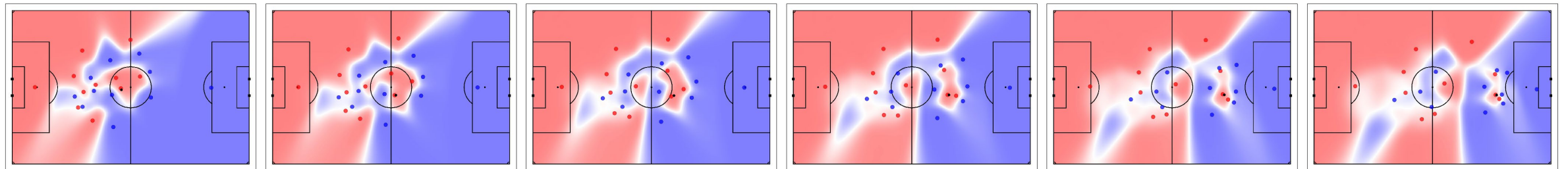


(C Gu and V De Silva, 2023)(<https://arxiv.org/abs/2303.13323>)

Results



REAL POSSESSION



Ghosting POSSESSION

Conclusion

► Contributions

Our deep generative ghosting model can be used as a benchmark for evaluating soccer players' performance on the pitch.

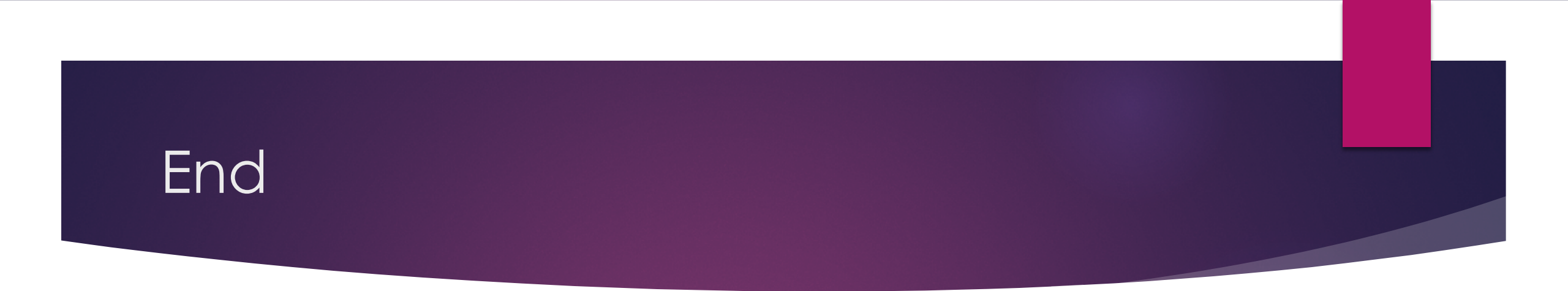
This model can be applied in analyzing team movement and individual player's ability of utilizing open space.

► Future Direction

The future research direction can be combining our method with expected goal model to quantify the value of each decision made within the game

References

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- ▶ Decroos, T., Bransen, L., Van Haaren, J., & Davis, J. (2021, January). VAEP: an objective approach to valuing on-the-ball actions in soccer. In Proceedings of the Twenty-Ninth International Conference on International Joint Conferences on Artificial Intelligence (pp. 4696-4700).
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- ▶ Rathke, A. (2017). An examination of expected goals and shot efficiency in soccer. Journal of Human Sport and Exercise, 12(2), 514-529.
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End

Thanks for listening !

Any question is welcomed

Further discussion can be made via email

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